Chemical Abundances in NGC 5024 (M53): A Mostly First Generation Globular Cluster

Owen Boberg, Eileen Friel, Enrico Vesperini
Indiana University

Motivation

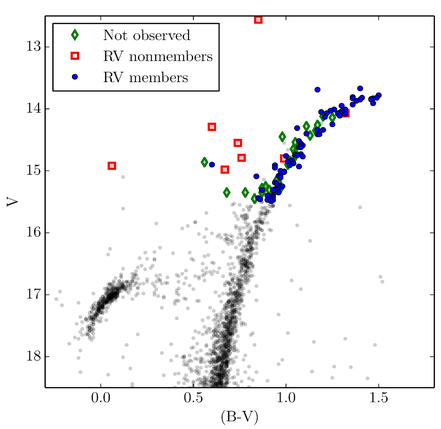
- M53 is a relatively massive, highly concentrated globular cluster
  - Underfills tidal radius
  - Morphology has implications for the degree of chemical enhancement and relative fraction of the multiple generations in M53
- Previous studies based on the morphology of M53's horizontal branch suggested that it might be composed primarily of first generation (FG) stars (Caloi et al. 2011)
- Majority of other GCs with multiple populations have been found to be dominated by the second generation (SG) stars
- Na and O abundances allow us to characterize the multiple populations in M53 and their relative fractions

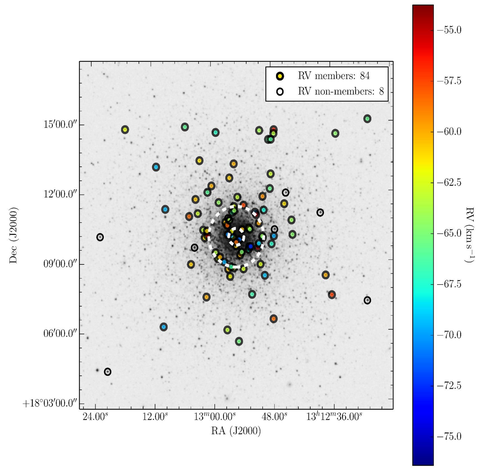
Sample Selection

- Selection criteria:
  - V < 15.5
  - Lie along the apparent RGB in V,B-V color magnitude diagram
  - Radial velocities consistent with the cluster:
    This study: -63.2 +/- 0.5 km s^{-1}
    Kimmig et al 2015: -62.8 +/- 0.3 km s^{-1}
  - R ~ 19,000 spectra were collected using Hydra at WIYN
  - λ range: 6063 - 6380 angstroms
  - Typical S/N ~ 80 - 120
  - Exposure times of ~4.5 hours per Hydra configuration

Analysis

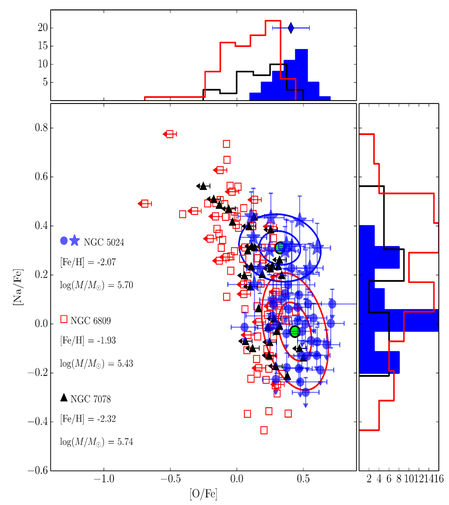
- Atmospheric parameters determined using Rey et al. 1995 B,V photometry and 2MASS J,H,K.
- Used relations in Alonso et al (1999), Carretta et al. (2004), Johnson et al. (2008), and Pilachowski et al. (1996)
- E(B-V) = 0.02, (m-M) = 16.32, M* = 0.8 M_{solar}
- MARCS models were generated based on the photometrically determined atmospheric parameters.
- Fe, Ca, Ti, and Ni abundances were determined from equivalent widths using the *abfind* task in MOOG.
- Na, O, and Ba abundances were determined by spectral synthesis using the *synth* task in MOOG.

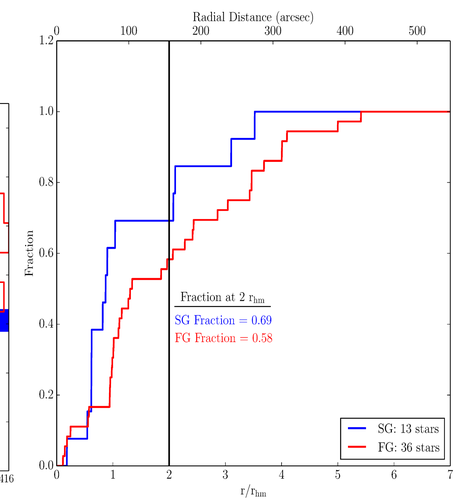




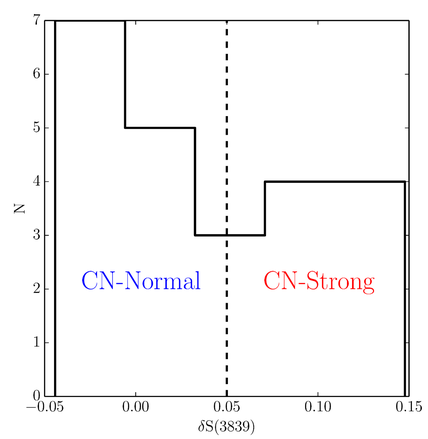
Left: V vs (B-V) CMD for M53. Data taken from Rey et al. (1998). The points are marked according to their cluster membership. **Right:** A 15' × 15' DSS image centered on M53. The RV members are marked with colored circles according to their individual radial velocities. The radial velocity non-members are marked with open black circles. The number of stars in each sample is given in the legend. The core and half light radii are shown as dashed white circles, respectively. The values for these radii were taken from the Harris 1996 catalog, version 2010.

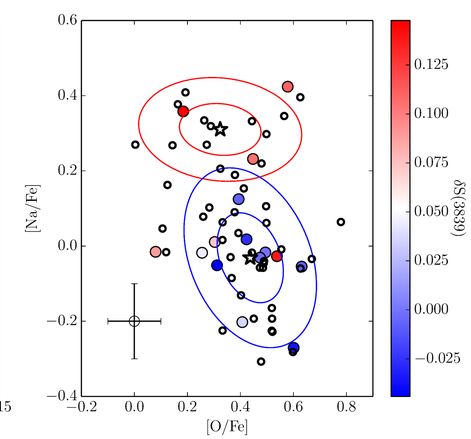
Na-O anti-correlation





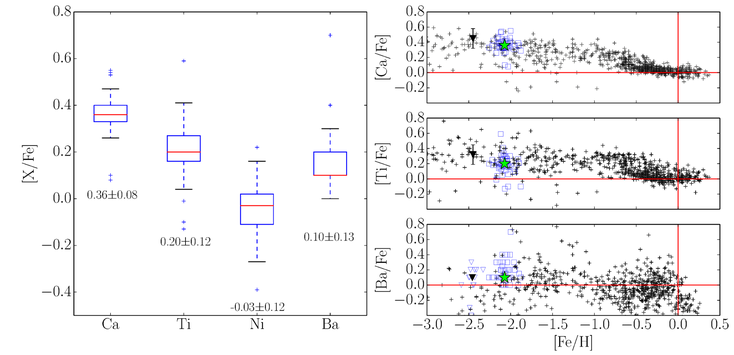
Left: [Na/Fe] vs [O/Fe] for M53 cluster members in our sample. The stars designated as **FG** are plotted as blue circles, **SG** stars are plotted as blue stars. The **FG** and **SG** stars were classified using a 2D Gaussian Mixture Model (GMM). The red and blue ellipses mark the 1 and 2σ levels of the FG and SG Gaussian components, respectively. Also plotted are the values for NGC 6809 and NGC 7078, taken from Carretta et al. 2009. **Right:** Cumulative radial distribution of the **FG** and **SG.** We find **13 FG** stars and **36 SG** stars.





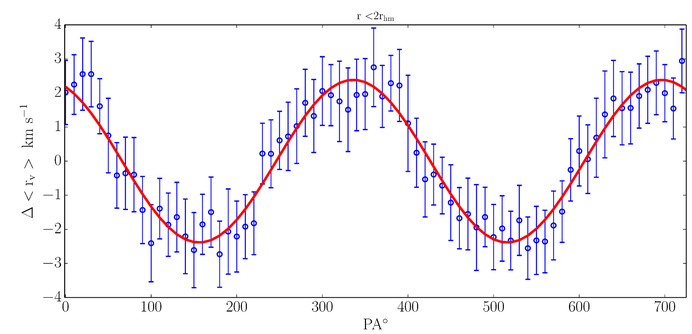
Left: CN band strength distribution in M53 from Martell et al. 2008. **Right:** The stars in our sample in common with Martell et al. 2008 are color coded according to their CN band strength on a plot of [Na/Fe] vs [O/Fe]. The open circle with error bars illustrates the typical errors in our measurements.

Ca, Ti, Ni, & Ba Abundances



Left: Box-and-whisker plot of the Ca, Ti, Ni and Ba abundances in M53. The red lines mark the median abundance of each species and the values are given below each box. **Right:** Ca, Ti, and Ba abundances of the stars in our sample plotted with a MW sample taken from Venn et al. (2004), plotted as black crosses. The open squares in each plot represent the individual abundances for each star in our sample, the solid green star marks the average of these individual abundances. The black triangle in each panel marks the average cluster abundance and error in the nearby GC NGC 5053 as measured in our previous study (Boberg et al. 2015).

Internal Rotation



- Rotation curve for stars within 2 half mass radii in M53
- Difference in mean velocity on either side of a line passing through the center of the cluster.
- The dividing line is rotated through the position angles shown on the x-axis and the difference is recalculated.
- The red line is a sine function fit to the data:

$$\Delta < r_v > = A_o sin(PA + \varphi)$$
$$\varphi = 270 - PA_0$$

- PA_0 is the position angle of the axis of rotation

Conclusions

- Cluster average [Fe/H] = -2.07, σ = 0.07 dex (From 57 stars)
- Ca, Ti, Ni, and Ba abundances consistent with Milky Way Halo population and other low metallicity globular clusters
- Using a 2D Gaussian mixture model, we classify two populations in the Na-O anti-correlation
  - **13 SG stars**
  - **36 FG stars**
- This is in contrast to most globular clusters which are dominated by the **SG**
  - **M53: 73% FG, 26% SG** (This work)
  - **NGC 6809: 20% FG, 77% SG** (Carretta et al. 2009)
  - **NGC 7078: 39% FG, 61% SG** (Carretta et al. 2009)
- The **SG** population is more centrally concentrated in M53
- Our findings are consistent with the HB morphology and high central concentration of M53
- There is not a completely one-to-one correlation between Na abundance and CN molecular band strength.
  - This has also been seen in M5 and 47 Tuc (Smith et al. 2013, Smith 2015)
- Our radial velocity analysis shows that there is internal rotation present in M53, which we will present in a future study.

References and Acknowledgments

Alonso, A., Arribas, S., & Martínez-Roger, C. 1999, A&AS, 140, 261
Boberg, O. M., Friel, E. D., & Vesperini, E. 2015, ApJ, 804, 109
Carretta, E., Bragaglia, A., Gratton, R. G., & Tosi, M. 2004, A&A, 422, 951
Carreta et al., 2009a A&A,505,117
Caloi, V., & D'Antona, F. 2011, MNRAS, 417, 228
Gustafsson, B., Edvardsson, B., Eriksson, K., et al. 2008, A&A, 486, 951 Harris, W. E. 1996, AJ, 112, 1487 (Cluster radii and Galactic positions)
Kimmig, B., Seth, A., Ivans, I. I., et al. 2015, AJ, 149, 53
Johnson et al. 2008, ApJ, 681, 1505
Martell, S. L., Smith, G. H., & Briley, M. M. 2008, PASP, 120, 7
Pilachowski, C. A., Sneden, C., & Kraft, R. P. 1996, AJ, 111, 1689
Rey, S.-C., Lee, Y.-W., Byun, Y.-I., & Chun, M.-S. 1998, AJ, 116, 1775
Smith, G. H. 2015, PASP, 127, 825
Smith, G. H., Modi, P. N., & Hamren, K. 2013, PASP, 125, 1287
Sneden, 1973, PhD Thesis, Univ. Texas at Austin (MOOG)
Venn, K. A., Irwin, M., Shetrone, M. D., et al. 2004, AJ, 128, 1177
Rey, S.-C., Lee, Y.-W., Byun, Y.-I., & Chun, M.-S. 1998, AJ, 116, 1775
Smith, G. H. 2015, PASP, 127, 825
Smith, G. H., Modi, P. N., & Hamren, K. 2013, PASP, 125, 1287
Sneden, 1973, PhD Thesis, Univ. Texas at Austin (MOOG)
Venn, K. A., Irwin, M., Shetrone, M. D., et al. 2004, AJ, 128, 1177

I am currently funded on an NSF Graduate Student Research Fellowship.

Email address: oboberg@indiana.edu
Website: oboberg.github.io